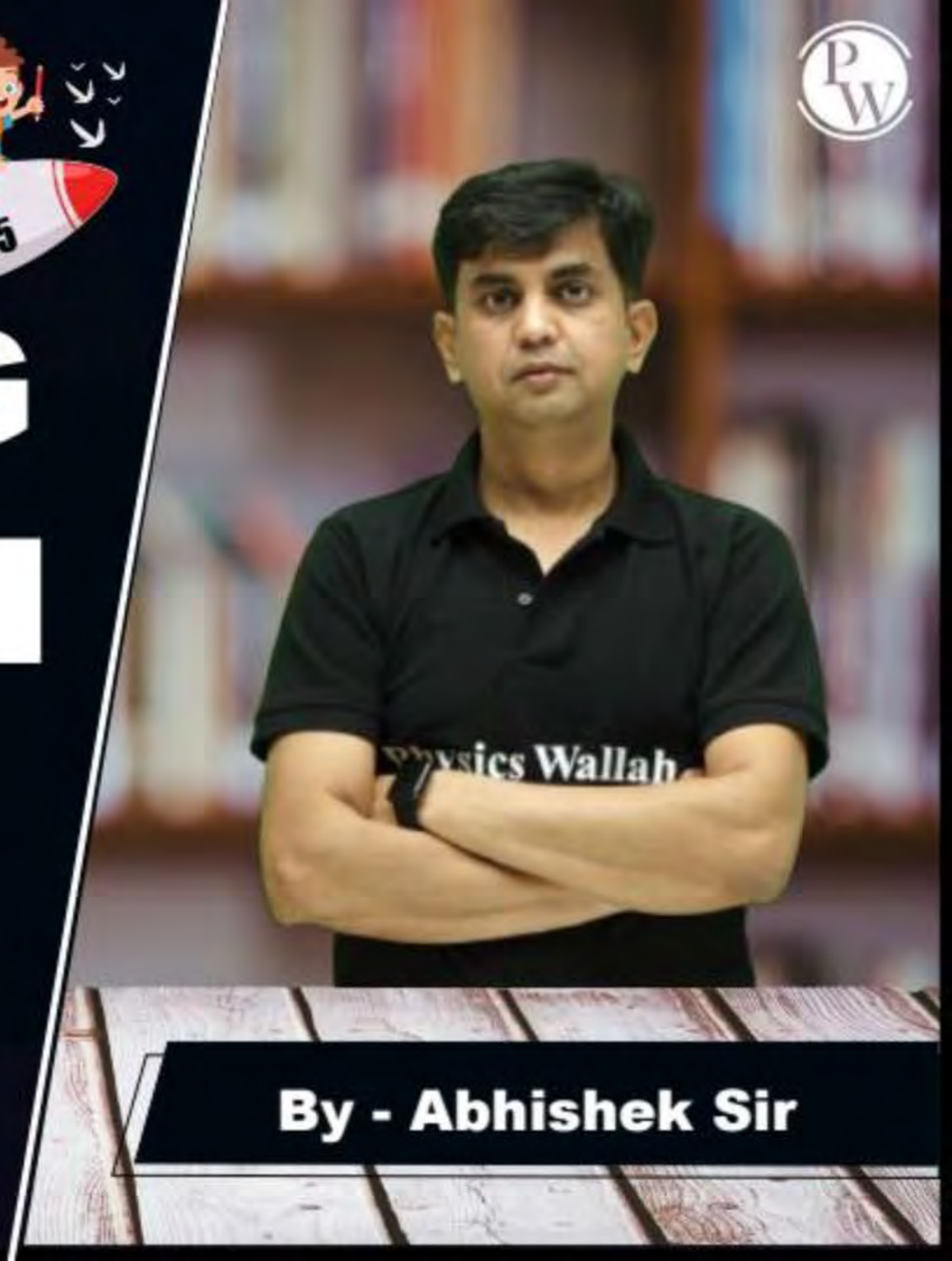
CS & IT ENGINEERING

Computer Network

Application Layer

Lecture No. - 04



Recap of Previous Lecture

Topic HTTP

Topics to be Covered

Topic Email

Topic SMTP

Topic FTP

ABOUT ME

Hello, I'm Abhishek

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
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[MSQ]

#Q. Suppose in a web browser, you click on the www.gate-2023.in URL. The browser cache is empty. The IP address for this URL is not cached in your local host, so a DNS lookup is triggered (by the local DNS server deployed on your local host) over the 3-tier DNS hierarchy in an iterative mode. No resource records are cached anywhere across all DNS servers.

Let RTT denote the round trip time between your local host and DNS servers in the DNS hierarchy. The round trip time between the local host and the web server hosting www.gate-2023.in is also equal to RTT. The HTML file associated with the URL is small enough to have negligible transmission time and negligible rendering time by your web browser, which references 10 equally small objects on the same web server.

Which of the following statements is/are CORRECT about the minimum elapsed time between clicking on the URL and your browser fully rendering it?

[GATE-2023, 2-Mark]

(A) 7 RTTs, in case of non-persistent HTTP with 5 parallel TCP connections. 10

(B) 5 RTTs, in case of persistent HTTP with pipelining.

(C) 9 RTTs, in case of non-persistent HTTP with 5 parallel TCP connections.

(D) 6 RTTs, in case of persistent HTTP with pipelining.

Domain : www. gate-2023 . in

Autho. N.S. TLD

Host

DNS Resolver (Iterative Mode) ⇒ 3 RTT

Ans: C & D

Non persistent HTTP with 5 parallel TCP connections

3RTT → DNS Resolver

2RTT → Base HTML File

2RTT → 5 small objects using 5 parallel TCP

2RTT → Remaining 5 small objects, using 5 parallel TCP

9RTT

Persistent HTTP with pipeline :-

3 RTT → DNS Resolver
1 RTT → TCP connection
1 RTT → Base HTML file
1 RTT → All 10 small object in pipeline

6 RTT

#Q. A user starts browsing a webpage hosted at a remote server. The browser opens a single TCP connection to fetch the entire webpage from the server. The webpage consists of a top-level index page with multiple embedded image objects. Assume that all caches (e.g., DNS cache, browser cache) are all initially empty. The following packets leave the user's computer in some order.

(i) HTTP GET request for the index page (base HTML file)
(ii) DNS request to resolve the web server's name to its IP address
(iii) HTTP GET request for an image object
(iv) TCP SYN to open a connection to the web server

Which one of the following is the CORRECT chronological order (earliest in time to latest) of the packets leaving the computer ?

IISC

[GATE-2024, Set-1, 1-Mark]

(ii) (iv) (i) (iii)

(A) (iv), (ii), (iii), (i)
(B) (ii), (iv), (iii), (i)
(C) (ii), (iv), (i), (iii)
(D) (iv), (ii), (i), (iii)

Ans: C

Topic : Email

3 major component of Email :

1. User Agent
2. Mail Server
3. SMTP (Protocol)
   + IMAP (POP)

Topic : User Agent

→ Mail Reader

→ Composing, editing, reading mail messages

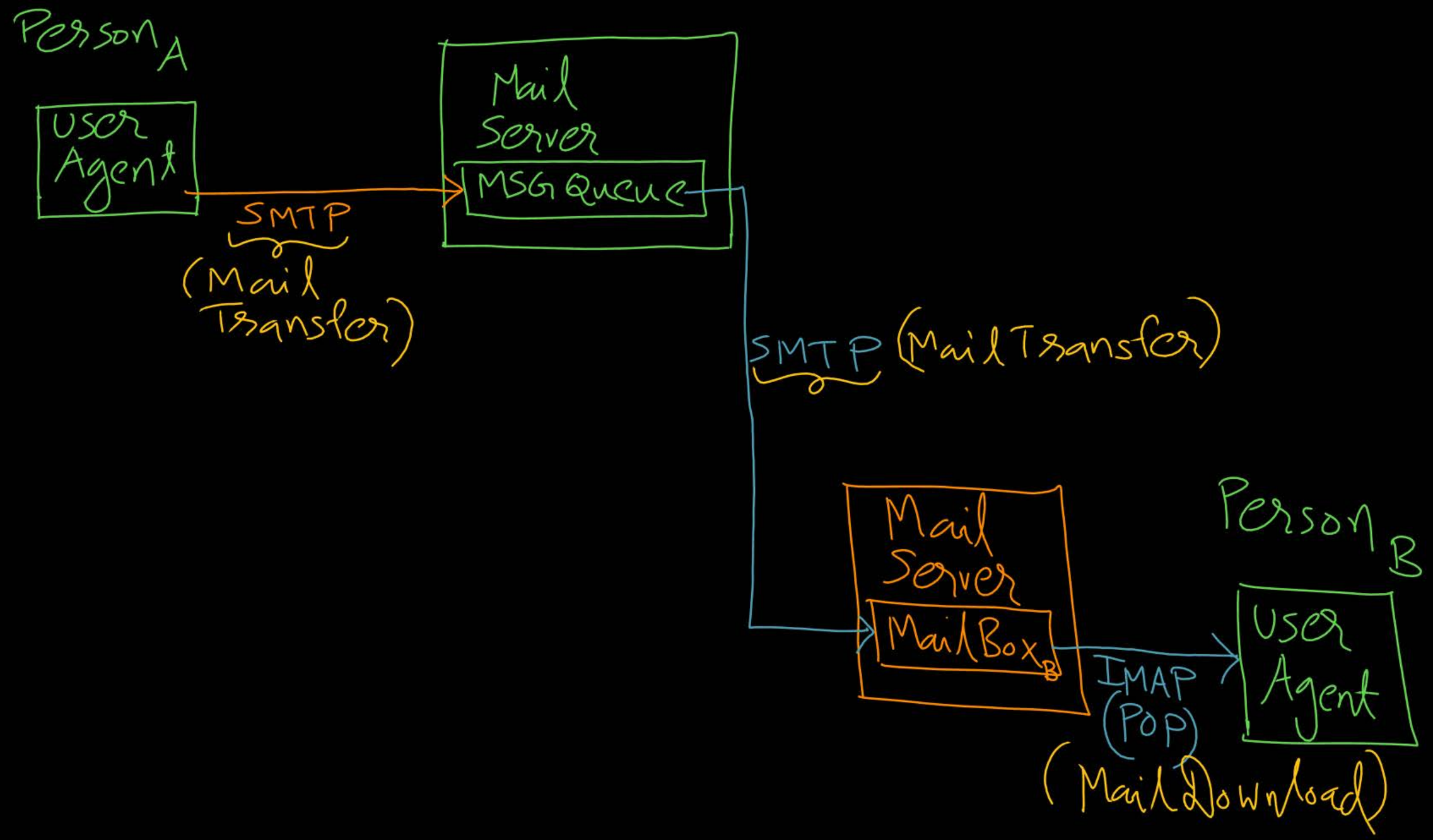
→ e.g Outlook

→ Outgoing, incoming messages stored on server

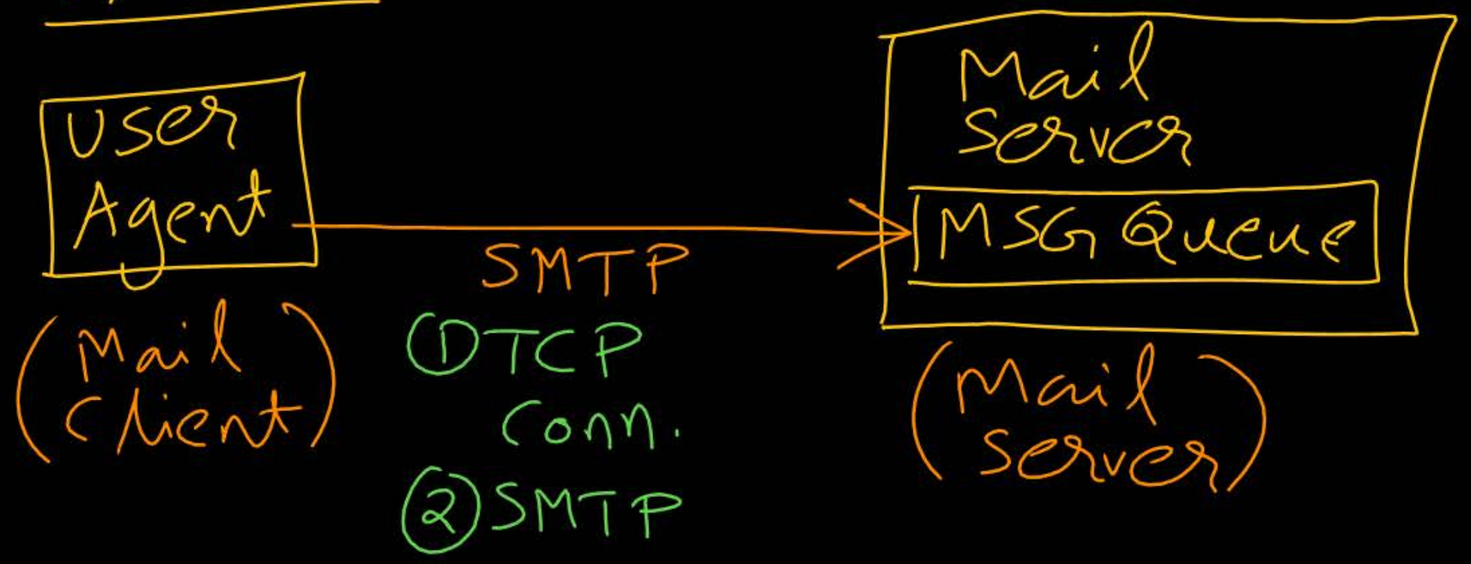
Topic : Mail Server

→ Mailbox : Contains incoming message for user

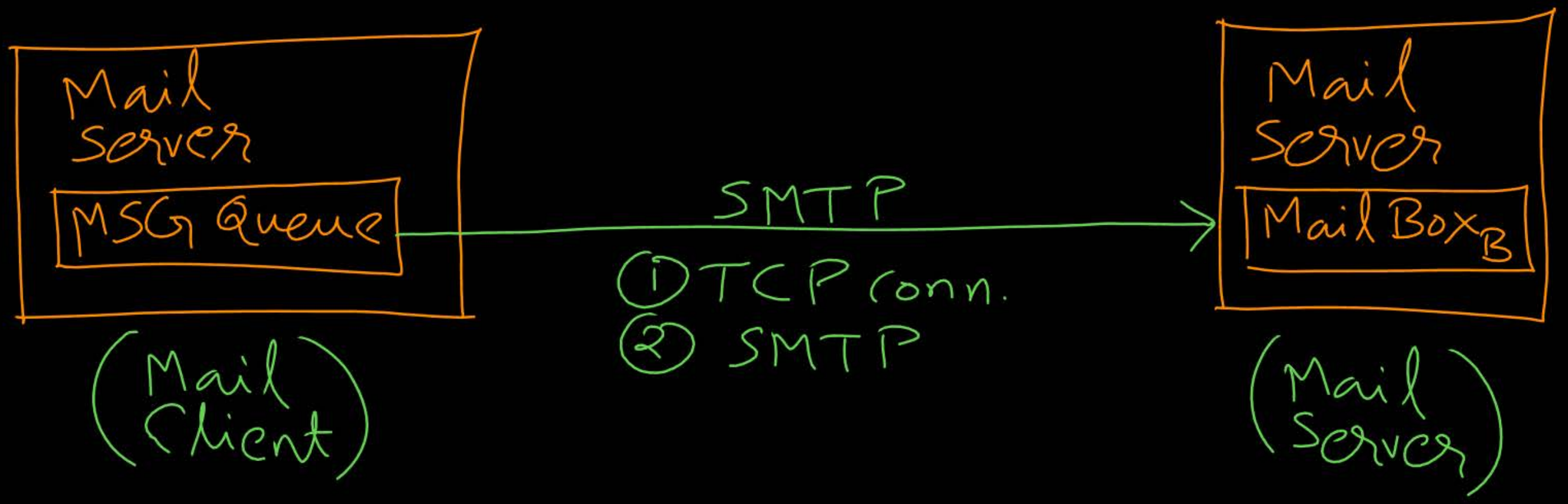
→ Message Queue : Outgoing (to be sent) mail messages


Person A
User Agent
SMTP
(Mail Transfer)
Mail Server
MSG Queue
SMTP (Mail Transfer)
Mail Server
MailBox B
IMAP
(POP)
(Mail Download)
Person B
User Agent

CASE I :-


User Agent
(Mail Client)
SMTP
① TCP Conn.
② SMTP
Mail Server
MSG Queue
(Mail Server)

CASE II :-


Mail Server
MSG Queue
(Mail Client)
SMTP
① TCP conn.
② SMTP
Mail Server
Mail Box$_B$
(Mail Server)

Topic : SMTP

→ SMTP : Simple Mail Transfer Protocol

→ Protocol between mail servers to send email messages (CASE II)

→ Client : Sending mail server

→ Server : Receiving mail server

Topic : SMTP

→ Application Layer 'Stateless' protocol

→ Uses TCP to reliably transfer email message
[from client mail server to server mail server]

SMTP ⟷ TCP ⟷ IP

→ Port No. 25

→ Sending server behave like client to receiving mail server

Topic : SMTP

→ Three phase of transfer :

1. SMTP Handshaking
2. SMTP transfer of messages
3. SMTP closure

① TCP conn. (3-way handshake)

② i) SMTP Handshake
ii) Transfer of mail
iii) Close
} mail 1

③ i) SMTP Handshake
ii) Transfer
iii) close
} mail 2

TCP conn. close
(4-way handshake)

Topic : SMTP

→ HTTP : Client Pull, SMTP : Client Push

→ SMTP uses persistent TCP connections

Topic : IMAP

→ Application Layer 'Stateful' protocol

→ IMAP : Internet Mail Access Protocol

→ Retrieving email from mail server
[Message stored on mailbox]

→ Earlier Protocol : POP (Post Office Protocol) (stateful)

POP vs IMAP

#Q. Which of the following protocol pairs can be used to send and retrieve e-mails (in that order)?

[GATE 2019, 1-Mark]

IIT-M

(A) IMAP, POP3

(B) SMTP, POP3

(C) SMTP, MIME

(D) IMAP, SMTP

Email :-

Send → SMTP

Retrieve → POP3/IMAP

Ans : B

#Q. Consider different activities related to email:

m1: Send an email from a mail client to a mail server : SMTP

m2: Download an email from mailbox server to a mail client : POP3/IMAP

m3: Checking email in a web browser : HTTP

Which is the application level protocol used in each activity?

[GATE 2011, 1-Mark]

IIT-M

(A) m1: HTTP m2: SMTP m3: POP

(B) m1: SMTP m2: FTP m3: HTTP

(C) m1: SMTP m2: POP m3: HTTP

(D) m1: POP m2: SMTP m3: IMAP

Ans: C

Topic : FTP

FTP ⟷ TCP ⟷ IP

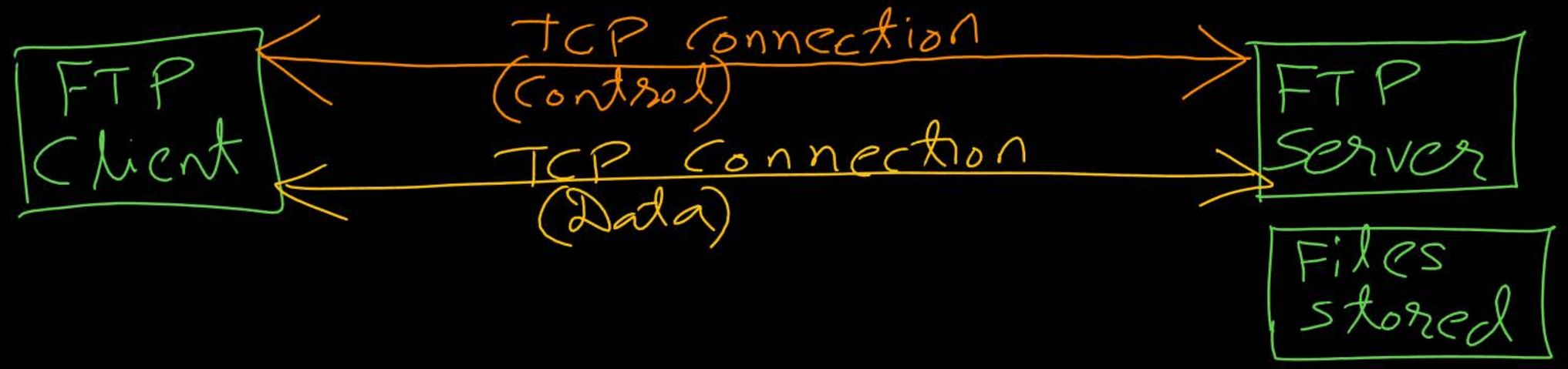
FTP : File Transfer Protocol

→ Reliable transfer of files [uses TCP as transport protocol]

→ Uses two TCP connection
[one for control and one for data]

→ Stateful protocol

→ Transfer any kind of file


FTP Client
TCP Connection (Control)
TCP Connection (Data)
FTP Server
Files stored

#Q. Which of the following is/are example(s) of stateful application layer protocols?

(i) HTTP stateless

(ii) FTP stateful

(iii) TCP stateful Transport Protocol

(iv) POP3 stateful

[GATE 2016, Set-1, 1-Mark]

IISC

(A) (i) and (ii) only

(B) (ii) and (iii) only

(C) (ii) and (iv) only

(D) (iv) only

Ans: C

Topic : FTP

→ FTP client contact FTP server at port 21 [opens TCP control connection] → control [Persistent TCP Conn.]

→ Client obtains authorization over TCP control connection

→ Client browses remote directory by sending commands over control connection

Topic : FTP

→ When FTP server receives a command for file transfer, the server opens a TCP data connection to client at port 20

→ After transferring one file, server closes TCP data connection } → Non persistent TCP conn.

→ TCP control connection remains open, to transfer another file

→ FTP server maintains "state" for connected clients

(1) FTP client make Request to download a file over TCP control connection

i) FTP server open TCP data connection

ii) Transfer file (server to client)

iii) close

(2) FTP client make Request to upload a file over TCP control connection

i) FTP client open TCP data connection

ii) Transfer file (client to server)

iii) close

#Q. In one of the pairs of protocols given below, both the protocols can use multiple TCP connections between the same client and the server. Which one is that?

[GATE 2015, Set-1, 1-Mark]

(A) HTTP, FTP → Non persistent HTTP IIT-K

(B) HTTP, ~~TELNET~~

(C) FTP, ~~SMTP~~

(D) HTTP, ~~SMTP~~

Ans: A

Topic : FTP Commands

→ ASCII (human-readable format)

→ Sent over control connection

1. USER username
2. PASS password
3. LIST : Return list of files in current directory
4. RETR filename : retrieves (gets) file from server
5. STOR filename : stores (puts) file onto server

Topic : FTP Responses

→ Status Code and Phrase

e.g "331 username ok, password required"

#Q. The transport layer protocols used for real time multimedia, file transfer, DNS and email, respectively are:

UDP TCP UDP TCP

[GATE 2013, 1-Mark]

IIT-B

(A) TCP, UDP, UDP and TCP

(B) UDP, TCP, TCP and UDP

(C) UDP, TCP, UDP and TCP

(D) TCP, UDP, TCP and UDP

Ans: C

#Q. Which one of the following protocols is NOT used to resolve one form of address to another one?

[GATE 2016, Set-1, 1-Mark]

IISc

(A) DNS [Host Name ⟷ Address]

(B) ARP [IP Add. ⟶ MAC Add.]

(C) DHCP IP Add. Allocation (Dynamic)

(D) RARP [MAC Add. ⟶ IP Add]

Ans: C

Topic : Stateful and stateless protocol

Transport Layer :-

Stateless : UDP

Stateful : TCP

Application Layer :-

Stateless : DNS, HTTP, SMTP

Stateful : FTP, IMAP (POP)

Topic : Syllabus

Concept of layering : OSI and TCP/IP Protocol Stacks;

Basics of packet, circuit and virtual circuit switching;

Data link layer : framing, error detection, Medium Access Control, Flow Control
Ethernet bridging;

CRC

[CSMA/CD]

Routing protocols : shortest path, flooding, distance vector and link state routing;

Topic : Syllabus

Fragmentation and IP addressing, IPv4, CIDR notation, Basics of IP support protocols (ARP, DHCP, ICMP), Network Address Translation (NAT);

Transport layer : flow control and congestion control, UDP, TCP, sockets;

Application layer protocols : DNS, SMTP, HTTP, FTP, Email.

2 mins Summary

Topic Email

Topic SMTP

Topic FTP

THANK - YOU